Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

01 What I built and why

1 Clarify — metric, period, population; asks if ambiguous

2 Verify definition — metric exists and its definition in registry + catalog

3 Route to a layer — Gold if it fits; Silver for detail; Bronze for raw audit

4 Query + evidence — read-only, bounded; one log per query, never overwritten

5 Test hypotheses — ≤3 per round, ≤3 rounds; stop when explained

6 Report — pattern ≠ cause; fixed PDF template, ≤10 pages

Read by every run, in Claude Code or Codex: AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

Gold
8 monthly tables · fast standard answers

Silver
10 cleaned tables · detail for investigations

Bronze
20 views over raw tables · audits

One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks first, as it did in all four evaluation questions that needed it ("which period?", "what counts as mobile?"), and it states every assumption it keeps.
- **Verifies the metric before computing it.** It checks that the metric exists and reads its approved, versioned definition and SQL in the metric registry, plus grains, joins and known gaps in the data catalog. Unsupported requests are declined, not improvised.
- **Maps the question to a layer for speed.** Gold answers standard monthly questions in milliseconds and is reconciled; Silver keeps per-user and per-event detail; Bronze is used only to audit raw values.
- **Evidence for every number.** Each query writes its SQL, layer, metric version and row count to its own log, so any figure in a report can be traced.
- **LLM-neutral by design.** Definitions, rules and guardrails live in files, not prompts, so Claude Code and Codex behave the same and a failure becomes a rule that every later run reads. Trade-off: no UI; quality depends on the agent following the skill.

Tools. Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and computed the answer key. Codex (GPT Terra 6.1, medium) ran 17 of 18 evaluation runs, testing the agent in a runtime that didn't build it. ChatGPT (Terra 5.6, later 6.1, medium) was the independent reviewer.

02 Where it breaks: before and after one improvement round

Failure mode	Where	Round 1	What changed	Round 2
Stating a cause the data can't prove; mixed comparison periods	GM report v1 (Q02)	Fail	Skill rules: pattern ≠ cause; exact periods; quantify or say "unknown"	Not seen again in 17 later runs
First month used as a normal baseline (subscriber base starts at zero)	Q07 cohorts	Fail	Rule: check for a start-up period before using it as a baseline	Pass
Same, in a trend description	Q11 funnel	Partial	(same rule)	Partial: rule didn't trigger
Harmless registry-version warning repeated	All runs	18 of 18	Change verified as docs-only and recorded	3 of 4
Overall, 18 questions (pass / partial / fail)		**15 / 1 / 2**		**16 / 1 / 1**

Hardest: comparisons that need context the data doesn't announce, such as a start-up month. **Strongest:** ambiguity and traps: it asked when it should have, declined the unanswerable question and caught both planted data-quality issues. Every answer was scored blind against values computed beforehand from the raw tables.

03 Running it for a 20-person product team

Question type	Example	Time to answer (observed)
Simple	MAU by month; US subscribers	2–3 minutes
Normal	Refund impact; funnel drop-off	5–6 minutes
Complex	MAU up 20% but revenue flat: why?	10–12 minutes, report included

- **Access and guardrails:** a read-only role on a warehouse replica; the query tool's limits stay on; sensitive columns kept out of the catalog (not built yet).
- **Cost:** not measured here; log tokens per run before setting a per-question budget.
- **Quality over time:** the 18 questions become a regression suite, rerun whenever the skill, registry or data model changes; track pass rate, clarification rate and query errors; review a weekly sample of real questions. Failures become rules, then a rerun.

04 With more time

- Make the start-up check apply whenever the first month looks unusual, not only when choosing a baseline (the remaining Q11 issue).
- Run the same questions in both runtimes for a direct Claude vs Codex comparison; add held-out questions.
- Measure cost per run; a short-answer mode for simple questions; a machine-readable catalog; a simple UI.